YANIS AKSAS

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EDUCATION

City University of Hong Kong — **Hong Kong, HK**
Exchange Program — Sep 2025 - Dec 2025
- Relevant coursework: Data Science, Foundations of Data Engineering

Polytechnique Montreal — **Montreal, QC**
Bachelor's in Mechanical Engineering - Dean's List — Sep 2020 - Dec 2025
- Relevant coursework: Python Programming, Data Structures, Numerical Modeling

WORK EXPERIENCE

CITY UNIVERSITY OF HONG KONG — **Hong Kong, HK**
Research Assistant — Oct 2025 - Dec 2025
- Conducted literature review across multi-agent LLM trading systems and hyperparameter optimization methods to justify a systematic trading-optimization proposal for a Hong Kong brokerage firm
- Co-designed the resulting 4-module workflow architecture (Strategist, Risk Manager, Memory Curator, Analyst) and multi-objective scoring framework

MMPROD — **Montreal, QC**
Software Developer — Sep 2024 - Feb 2025
- Built an offline desktop translation app for MMPROD using MarianMT, engineering a PDF and DOCX pipeline that preserved layout, tables, and images across 200+ page documents

FENIX — **Montpellier, France**
Software Engineer Intern — May 2024 - Aug 2024
- Designed and executed web scraping pipeline using Python and pandas to collect 9,000+ prospect contacts across France with geographic segmentation, building structured Excel Dataset for targeted marketing campaigns
- Redesigned company website using JavaScript with UX/UI optimization and implemented analytics tracking system, measuring 45% traffic increase and 85% conversion improvement post-launch

LEADERSHIP EXPERIENCE

POLY FINANCES — **Montreal, QC**
Software Developer — Jan 2025 - May 2025
- Developed and maintained a portfolio visualization tool and job board (FinXplore) for a $100K+ student-managed fund using Python and Angular
- Conducted technical analysis and presented investment strategies to industry professionals

DATA RANK — **Montreal, QC**
CTO — Dec 2024 - Present
- Co-founded DataRank as CTO, an invitation-based platform that scores a company's data maturity through AI-assisted questionnaires and interviews and generates a PDF report with prioritized recommendations for internal, client, or pre-audit review
- Built the platform's AI evaluation pipeline on an Express/Supabase backend with Stripe billing, including GPT-4 agents for pattern detection, benchmark comparison, and risk assessment

PROJECT EXPERIENCE

Financial Sentiment Analysis (Python) — **Montreal, QC**
Jun 2025 - Aug 2025
- Built a 4-model NLP sentiment pipeline (FinBERT, BERT, VADER, TextBlob) scoring financial news for 12 large-cap companies, finding only 37% label agreement between FinBERT and general-purpose BERT on the same headlines
- Tested FinBERT sentiment against next-day forward returns pooled across 141 ticker-day observations, finding no statistically significant relationship ($r \approx 0.000$), a well-powered null result consistent with market efficiency

Energy Production Optimization Model (Python) — **Montreal, QC**
Sep 2024 - May 2025
- Built the network optimization module for HarmoniQ, a multi-source grid platform for Hydro-Québec, using PyPSA for grid topology and AC/DC power flow analysis, with linear programming (HiGHS solver) dispatch minimizing cost and CO2 emissions across 500,000+ hourly data points
- Co-built an interactive dashboard visualizing real-time network status, scenario planning, and dispatch results across hydro, wind, solar, nuclear, and thermal sources, enabling data-driven decisions for grid operators

SKILLS & INTERESTS

Programming: Python (NumPy, Pandas, TensorFlow, PyTorch), C++, R, SQL, MATLAB, JavaScript, Claude Code
Certificates: Machine Learning (Stanford), Financial Engineering & Risk Management (Columbia)